

Hamza Arfin

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ABOUT

Full-stack developer experienced in building web applications using Python, Flask, and MySQL. Skilled in REST API development, database design, and backend systems with projects in recommendation systems, NLP, and data-driven applications.

SKILLS

Languages: Python, C, C++, JavaScript, SQL
Web: Flask, HTML, CSS, REST APIs, Backend Development, Jekyll
ML/NLP: NumPy, Pandas, Scikit-learn, BERT, Transformers, PyTorch
Database: MySQL, SQLAlchemy
Tools: Git, GitHub
Concepts: Data Structures, Algorithms, CRUD, API Integration

EXPERIENCE

FLAme NLP Research Lab, IIIT Delhi

July 2025 – September 2025

Undergraduate Research Intern

New Delhi, India

- Developed and deployed the official lab website (FLAme NLP Lab) using **Jekyll**
- Built a scalable static website with modular components for managing research updates, publications, and team information
- Structured content using templates and data files to enable easy updates without modifying core code
- Improved usability, performance, and maintainability of the lab's digital presence

PROJECTS

BERT Sarcasm Detection Classifier | GitHub

May 2026

- Fine-tuned **bert-base-uncased** on the News Headlines sarcasm dataset using stratified split and weighted loss
- Optimized classification threshold to **0.3965**, reaching **93.77% F1**, **94.06% accuracy**, and **0.983 ROC-AUC**
- Built **FastAPI** inference endpoint and browser UI returning predictions, confidence scores, explanations, and tags
- Generated reproducible evaluation artifacts including confusion matrix, ROC/PR curves, threshold-F1 curve, and training logs

Hybrid Movie Recommendation System | GitHub

March 2026

- Built a recommendation platform using **Python, Flask, HTML, CSS, JavaScript**
- Implemented hybrid recommender using **KNN collaborative filtering** on the **MovieLens dataset (100k+ ratings)**
- Integrated **TMDB API** and built REST endpoints serving movie metadata and posters

Streamly – Streaming Aggregator | GitHub

November 2025

- Developed a streaming search platform using **Flask** and the **TMDB API**
- Implemented real-time search suggestions and optimized API requests with rate limiting

Flask Hospital Management System

August 2024

- Built hospital management web app using **Flask, MySQL, SQLAlchemy**
- Designed relational database schema managing patient, doctor, and appointment records across multiple tables

Python–MySQL Hospital Database System | GitHub

October 2023

- Built command-line hospital record system using **Python** and **MySQL** with automatic database and table setup
- Implemented CRUD workflows for doctors, nurses, workers, and patients, including patient admission and discharge handling
- Used parameterized SQL queries, input validation, environment-based credentials, and tabular console output for safer operations

EDUCATION

Jamia Hamdard University <i>Bachelor of Technology in Computer Science and Engineering</i> • Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering • Focused on web development, backend systems, and database-driven applications	2023 – 2027 New Delhi, India
Dev Samaj Modern School - 2 <i>Senior Secondary School</i>	2021 – 2023 New Delhi, India
Dev Samaj Modern School - 2 <i>Secondary School</i>	2019 – 2021 New Delhi, India